

THEME 7 · B2.1

Life in Nature & Global Problems

Natural disasters and their impact on nature

Natural disasters and their impact on humans

Impact of technology on global issues (AI, data, healthcare)

Warm-up

Look at the title: Life in Nature & Global Problems. What do you think we will talk about in this theme? Tell your partner two ideas.

VOCABULARY

New words

20 target words

natural disaster

n. phrase



doğal afet

The earthquake was one of the worst natural disasters in the region's history.

earthquake

n.



deprem

The earthquake destroyed dozens of old buildings in the city centre.

flood

n.



sel

Heavy rain caused a flood that damaged the whole valley.

wildfire

n.



orman yangını

The wildfire spread across the hillside within hours.

drought

n.



kuraklık

The drought has made it difficult for farmers to grow crops this year.

landslide

n.



toprak kayması

Heavy rain triggered a landslide that blocked the mountain road.

evacuate

v.



tahliye etmek

Authorities evacuated the coastal town before the storm arrived.

aftermath

n.



bir olayın ardından yaşanan dönem

In the aftermath of the flood, hundreds of families needed shelter.

devastate

v.



harap etmek, yıkıma uğratmak

The wildfire devastated thousands of hectares of forest.

ecosystem

n.



ekosistem

The flood changed the river's ecosystem for years afterwards.

displacement

n.



yerinden edilme, göçe zorlanma

The earthquake caused the displacement of thousands of residents.

resilience

n.



dayanıklılık, toparlanma gücü

The town's resilience impressed the international relief teams.

relief effort

n. phrase



yardım çalışması

Volunteers joined the relief effort within days of the flood.

artificial intelligence (AI)

n. phrase



yapay zeka

Artificial intelligence can now analyse satellite images to spot early signs of a wildfire.

predict

v.



tahmin etmek, öngörmek

Scientists used weather data to predict the storm three days in advance.

algorithm

n.



algoritma

The app's algorithm compares rainfall data with river levels to warn of floods.

data analysis

n. phrase



veri analizi

Data analysis helped hospitals prepare for the number of patients after the earthquake.

early warning system

n. phrase



erken uyarı sistemi

The early warning system gave residents twenty minutes to leave the area.

sustainable

adj.



sürdürülebilir

Rebuilding with sustainable materials can help the town recover safely.

vulnerable

adj.



kırılğan, savunmasız

Coastal communities are particularly vulnerable to rising sea levels.

Quick game · What's the word?

“erken uyarı sistemi”

a drought

b vulnerable

c early warning system

Quick game · What's the word?

“**algoritma**”

a algorithm

b flood

c relief effort

Quick game · What's the word?

“dayanıklılık, toparlanma gücü”

a natural disaster

b data analysis

c resilience

Quick game · What's the word?

“sürdürülebilir”

a artificial intelligence
(AI)

b resilience

c sustainable

Quick game · What's the word?

“bir olayın ardından yaşanan dönem”

a predict

b aftermath

c vulnerable

Quick game · What's the word?

“tahliye etmek”

a evacuate

b landslide

c ecosystem

GRAMMAR

Defining Relative Clauses (who / which / that)

Notice

Scientists **who study earthquakes** can now predict some tremors more accurately.

The flood **that destroyed the village** changed the way the government plans for disasters.

This is the kind of technology **which helps doctors diagnose diseases faster**.

People **who live near the coast** are usually the most vulnerable to storms.

In each sentence, does the relative clause give essential information about the noun before it, or could we remove it without losing important meaning? Which word introduces each clause?

Complete the rule

Look at the Notice sentences again. Complete the rule with the words in the box.

who**which****that****no commas**

Complete the rule

We use _____ or _____ for people.

✓ who / that

Complete the rule

We use _____ or _____ for things.

✓ which / that

Complete the rule

**Defining relative clauses give essential information, so we use
_____ around them.**

✓ no commas

The rule

A defining relative clause gives essential information about the noun in the main clause; without it, the sentence's meaning would be incomplete or unclear. Use who/that for people and which/that for things. No commas are used with defining relative clauses.

TR Defining (tanımlayıcı) relative clause, ana cümledeki isim hakkında zorunlu/temel bilgi verir; bu bilgi olmadan cümlelerin anlamı eksik veya belirsiz kalır. İnsanlar için who/that, nesneler için which/that kullanılır. Bu tür cümlelerde virgül KULLANILMAZ.

When? We use defining relative clauses when we need to make clear exactly which person or thing we are talking about — for example, identifying one specific disaster among many.

TR Hangi kişi ya da nesneden bahsettiğimizi netleştirmek istediğimizde defining relative clause kullanırız — örneğin birçok deprem arasından belirli birini tanımlarken.

Build sentences · Example

Scientists study earthquakes.

+

They can now predict some tremors.

Scientists who study earthquakes can now predict some tremors.

Build sentences

The disaster destroyed hundreds of homes. + It happened last spring.

The disaster that happened last spring destroyed hundreds of homes.

Build sentences

An algorithm analyses weather data. +

It can warn people hours before a storm.

An algorithm that analyses weather data can warn people hours before a storm.

Build sentences

The volunteers helped flood victims.

+

They worked day and night.

The volunteers who worked day and night helped flood victims.

Build sentences

This is the app.

+

It predicts wildfires using AI.

This is the app which predicts wildfires using AI.

Build sentences

? / the scientists / develop early warning systems / do you know them

Do you know the scientists who develop early warning systems?

Your turn

Write four sentences about natural disasters or technology, each with a defining relative clause (who/which/that).

Scientists who...

The disaster that...

A system which...

People who...

Form

POSITIVE

Scientists who study earthquakes work at the university.
The app that predicts floods is free.

NEGATIVE

People who don't prepare for disasters are more at risk.
This isn't the system which we tested last year.

QUESTION

Is this the technology that helps predict wildfires?
Do you know the scientist who developed this algorithm?

SHORT ANSWERS

Yes, it is. / No, it isn't.
Yes, I do. / No, I don't.

Error detective

The scientist, who developed the app, works at our university.

a Correct

b Incorrect

✓ **The scientist who developed the app works at our university.**

Error detective

This is the village which it was destroyed by the flood.

a Correct

b Incorrect

✓ **This is the village which was destroyed by the flood.**

A • Choose the correct option.

The volunteers _____ arrived first saved many lives.

a who

b which

c whom

A • Choose the correct option.

This is the drone _____ films the flooded areas from above.

a who

b which

c whose

A • Choose the correct option.

People _____ live in coastal areas face a higher flood risk.

a which

b who

c whom

B · Complete with **who / which / that**.

The scientist _____ built the early warning app won an award.

✓ **who / that**

B · Complete with **who / which / that**.

This is the technology _____ helps doctors predict disease outbreaks.

✓ that / which

B · Complete with **who / which / that**.

The town _____ was hit by the earthquake is slowly rebuilding.

✓ **that / which**

C · Judge the sentence.

The engineer, who designed the system, is famous.

a Correct

b Incorrect

✓ **The engineer who designed the system is famous.**

C · Judge the sentence.

This is the app that predicts wildfires.

a Correct

b Incorrect

Dialogue · Complete with that/which.

A: Have you seen the video _____ shows the flood damage?

✓ **that / which**

Dialogue · Complete with that/which.

B: Yes! It's the one _____ went viral last week.

✓ **that / which**

Dialogue · Reorder the sentence.

predicts / the app / floods / which / is very accurate

floods

the

is

which

accurate

app

very

predicts

•

✓ **The app which predicts floods is very accurate.**

GRAMMAR

Non-Defining Relative Clauses (who / which, with commas)

Notice

The 2023 earthquake, **which affected millions of people**, changed disaster policy across the region.

Dr. Aylin Kaya, **who leads the early warning project**, spoke at the conference yesterday.

Wildfires, **which are becoming more frequent because of climate change**, threaten entire ecosystems.

My uncle, **who works for a relief organisation**, has helped after three major floods.

Is the information in the relative clause essential, or is it extra/additional information? What punctuation marks it off from the rest of the sentence? Could you use 'that' instead of 'who/which' here?

Complete the rule

Complete the rule with the words in the box.

extra/additional

commas

cannot use 'that'

Complete the rule

Non-defining relative clauses give _____ information — the sentence still makes sense without them.

✓ extra/additional

Complete the rule

We separate a non-defining clause from the rest of the sentence with _____.

✓ commas

Complete the rule

In non-defining clauses, we _____ .

✓ cannot use 'that'

The rule

A non-defining relative clause adds extra information about the noun; the sentence still makes complete sense without it. Commas are used before and after the clause, and 'that' is NOT used in non-defining clauses.

TR Non-defining (tanımlayıcı olmayan) relative clause, ana cümledeki isim hakkında EK/fazladan bilgi verir; bu bilgi çıkarılsa bile cümle hâlâ anlamlıdır. Cümlelerin başında ve sonunda VİRGÜL kullanılır ve bu tür cümlelerde 'that' KULLANILMAZ.

When? We use non-defining relative clauses to add an interesting or extra detail about a noun whose identity is already clear (proper names, unique people or places).

TR Zaten kim/ne olduğu belli olan bir isim (özel isimler, tek olan yer/kişiler) hakkında ilginç ya da ek bir detay eklemek istediğimizde non-defining relative clause kullanırız.

Build sentences · Example

The Kobe earthquake which happened in 1995 killed thousands of people.

The Kobe earthquake, which happened in 1995, killed thousands of people.

Build sentences

Professor Demir who studies climate change gave a lecture on droughts.

Professor Demir, who studies climate change, gave a lecture on droughts.

Build sentences

The Amazon rainforest which produces a huge amount of oxygen is losing trees every year.

The Amazon rainforest, which produces a huge amount of oxygen, is losing trees every year.

Build sentences

This app which was created by students won a technology prize.

This app, which was created by students, won a technology prize.

Build sentences

My cousin who volunteers for disaster relief is travelling to the flood zone.

My cousin, who volunteers for disaster relief, is travelling to the flood zone.

Your turn

Write three sentences about people or places you know, adding extra (non-defining) information with who/which and commas.

My..., who..., ...

..., which..., is/was...

Form

POSITIVE

The 2023 earthquake, which affected millions, changed disaster policy.

Dr. Kaya, who leads the project, spoke yesterday.

NEGATIVE

The report, which wasn't published until March, surprised experts.

My uncle, who doesn't usually travel far, went to help.

QUESTION

Have you read about the Amazon, which is losing trees every year?

Do you know Dr. Kaya, who developed the warning system?

SHORT ANSWERS

Yes, I have. / No, I haven't.

Yes, I do. / No, I don't.

Error detective

The Amazon rainforest, that produces oxygen, is shrinking.

a Correct

b Incorrect

✓ **The Amazon rainforest, which produces oxygen, is shrinking.**

Error detective

My uncle who works for a relief organisation is here.

a Correct

b Incorrect

✓ **My uncle, who works for a relief organisation, is here.**

A • Choose the correct option.

Mount Everest, _____ is the world's highest mountain, attracts thousands of climbers.

a that

b which

c who

A • Choose the correct option.

My sister, _____ works as a nurse, helped after the earthquake.

a who

b that

c which

A · Choose the correct option.

Which sentence is punctuated correctly?

a The scientist, who developed the app won an award.

b The scientist who developed the app, won an award.

c The scientist, who developed the app, won an award.

B · Complete with who / which.

The flood, _____ destroyed hundreds of homes, happened in April.

✓ which

B · Complete with **who** / **which**.

Professor Aydın, _____ leads our research team, is studying drought patterns.

✓ **who**

C · Judge the sentence.

The report that we published last year, was very detailed.

a Correct

b Incorrect

✓ **The report that we published last year was very detailed.**

C · Judge the sentence.

Türkiye, which lies on two continents, experiences frequent earthquakes.

a Correct

b Incorrect

Dialogue · Complete with who/which.

A: Have you met Dr. Kaya, _____ leads the AI disaster project?

✓ **who**

Dialogue · Complete with who/which.

B: Not yet, but I read her report, _____ was published last month.

✓ **which**

GRAMMAR

Omitting the Relative Pronoun (Object Relative Clauses)

Notice

The report **(that)** scientists published last year warned about rising sea levels.

The app **(which)** my brother uses tracks earthquake activity in real time.

I still remember the storm **(that)** we experienced two years ago.

The volunteers **(who)** the organisation trained worked in the flood zone.

In each sentence, is the person/thing in the relative clause the subject or the object of the verb inside that clause? Which word could be left out without changing the meaning?

Complete the rule

Complete the rule with the words in the box.

object

subject

cannot be omitted

Complete the rule

When the relative pronoun is the _____ of the relative clause verb (the clause already has its own subject), it can often be omitted.

✓ **object**

Complete the rule

When the relative pronoun is the _____ of the relative clause verb, it must stay.

✓ **subject**

Complete the rule

A relative pronoun that is the subject of the clause _____.

✓ **cannot be omitted**

The rule

The relative pronoun (that/which/who) can usually be omitted when it is the OBJECT of the relative clause (the clause has its own subject). It cannot be omitted when it is the SUBJECT of the clause. This rule only applies to defining relative clauses.

TR Relative pronoun (that/which/who), relative clause içinde NESNE görevindeyse (clause'un kendi öznesi varsa) genellikle atlanabilir. Ancak relative pronoun clause'un ÖZNESİ ise atlanamaz. Bu kural sadece defining relative clause'lar için geçerlidir.

When? We often omit the object relative pronoun in speech and everyday writing to make sentences sound more natural and less formal.

TR Konuşma dilinde ve günlük yazıda cümleyi daha akıcı hale getirmek için nesne konumundaki relative pronoun'u atlarız.

Build sentences · Example

The scientist that we interviewed studies volcanoes. (object)

The scientist we interviewed studies volcanoes.

Build sentences

The data that the sensors collect helps predict earthquakes. (object)

The data the sensors collect helps predict earthquakes.

Build sentences

The engineers who designed the app are from Izmir. (subject)

cannot omit — 'who' is the subject: The engineers who designed the app are from Izmir.

Build sentences

The house that the storm destroyed was empty. (object)

The house the storm destroyed was empty.

Build sentences

The system which predicts floods was built by students. (subject)

cannot omit — 'which' is the subject: The system which predicts floods was built by students.

Your turn

Write two sentences with an object relative clause where you can omit the pronoun, and one sentence with a subject relative clause where you cannot.

The ... I ...

The ... that/which ... (subject) ...

Form

POSITIVE

The report (that) we read
was alarming.

The scientist who studies
volcanoes gave a talk.

NEGATIVE

The data (that) we
collected wasn't complete.

The system which predicts
floods doesn't always
work.

QUESTION

Is this the app (that) you
downloaded?

Is this the scientist who
developed the app?

SHORT ANSWERS

Yes, it is. / No, it isn't.

Yes, it is. / No, it isn't.

Error detective

The scientist studies volcanoes gave a talk.

a Correct

b Incorrect

✓ **The scientist who studies volcanoes gave a talk.**

Error detective

This is the report which, I read yesterday.

a Correct

b Incorrect

✓ **This is the report (which) I read yesterday.**

A • Choose the correct option.

Which sentence **CANNOT** drop the relative pronoun?

a The house (that) the flood destroyed is empty.

b The scientist who studies volcanoes gave a talk.

c The report (that) we published was long.

A • Choose the correct option.

The app _____ my brother downloaded tracks earthquakes.
(most natural, informal)

a that

b — (no word needed)

c whom

A • Choose the correct option.

The volunteers _____ arrived first started the rescue.

a (nothing)

b who

c —

B · Complete with the correct form.

The data the sensors _____ (collect) helps scientists predict storms.

✓ collect

B · Complete with the correct form.

This is the algorithm _____ engineers designed for flood prediction. (formal style, add the pronoun)

✓ that / which

C · Judge the sentence.

The engineers designed the app won an award.

a Correct

b Incorrect

✓ **The engineers who designed the app won an award.**

C · Judge the sentence.

The report we read yesterday was alarming.

a Correct

b Incorrect

Dialogue · Complete with that/which.

A: Is this the data _____ the AI system uses? (formal)

✓ that / which

Dialogue · Reorder the sentence.

destroyed / the flood / the house / was empty

flood

empty

was

destroyed

house

the

the

•

✓ **The house the flood destroyed was empty.**

GRAMMAR

Relative Clauses with 'whose' (Possession)

Notice

The scientist **whose research predicted the earthquake** received an award.

We interviewed a family **whose house was destroyed by the flood**.

This is the village **whose water supply was affected by the drought**.

The company **whose app warns people about wildfires** is expanding across the country.

What does 'whose' show — possession, location or time? Can 'whose' be used only for people, or for people and things too?

Complete the rule

Complete the rule with the words in the box.

possession

people and things

noun

Complete the rule

'Whose' shows _____ — something belongs to someone or something.

✓ **possession**

Complete the rule

'Whose' can be used for _____, unlike who/which.

✓ **people and things**

Complete the rule

After 'whose' we always need a _____ (whose + noun).

✓ **noun**

The rule

'Whose' is a possessive relative pronoun used to show that something belongs to a person or thing already mentioned. It can be used for people AND things/places, and it is always followed by a noun (whose + noun).

TR 'Whose', bir kişi ya da şeye ait olan bir başka isim hakkında bilgi vermek için kullanılan iyelik bildiren relative pronoun'dur. Hem insanlar hem de nesneler/yerler için kullanılabilir ve kendisinden sonra mutlaka bir isim gelir (whose + isim).

When? We use 'whose' to introduce a feature that belongs to a person or place — their house, family, system, and so on.

TR Bir kişinin ya da yerin sahip olduğu bir özelliği (evi, ailesi, sistemi vb.) tanıtırken 'whose' kullanırız.

Build sentences · Example

We interviewed a family.

+

Their house was destroyed.

We interviewed a family whose house was destroyed.

Build sentences

The scientist won an award.

+

Her research predicted the earthquake.

The scientist whose research predicted the earthquake won an award.

Build sentences

The village lost its water supply.

+

Its wells dried up during the drought.

The village whose wells dried up during the drought lost its water supply.

Build sentences

The company is expanding. + **Its app warns people about wildfires.**

The company whose app warns people about wildfires is expanding.

Build sentences

I spoke to a boy.

+

His school was flooded last year.

I spoke to a boy whose school was flooded last year.

Your turn

Write three sentences with 'whose', describing people, places or organisations connected to natural disasters or technology.

The scientist whose...

A family whose...

The city whose...

Form

POSITIVE

The scientist whose research predicted the earthquake won an award.
The village whose wells dried up needs help.

NEGATIVE

The family whose house wasn't damaged still helped their neighbours.
This isn't the company whose app failed.

QUESTION

Is this the scientist whose research predicted the earthquake?
Do you know the family whose house was destroyed?

SHORT ANSWERS

Yes, it is. / No, it isn't.
Yes, I do. / No, I don't.

Error detective

The scientist who's research predicted the earthquake won an award.

a Correct

b Incorrect

✓ **The scientist whose research predicted the earthquake won an award.**

Error detective

The village whose it lost its water supply needs help.

a Correct

b Incorrect

✓ **The village whose water supply was lost needs help.**

A • Choose the correct option.

The engineer _____ app predicts wildfires works at our university.

a who

b whose

c which

A • Choose the correct option.

This is the town _____ water supply was affected by the drought.

a who

b whose

c which

A · Choose the correct option.

Choose the correct sentence.

a The scientist who's
research is famous
spoke today.

b The scientist whose
research is famous
spoke today.

c The scientist which
research is famous
spoke today.

B · Complete with whose.

We met a family _____ house had been flooded twice.

✓ whose

B · Complete with whose.

The company _____ app won the innovation prize is based in Ankara.

✓ whose

C · Judge the sentence.

The scientist who's research predicted the flood is famous now.

a Correct

b Incorrect

✓ **The scientist whose research predicted the flood is famous now.**

C · Judge the sentence.

This is the village whose wells dried up during the drought.

a Correct

b Incorrect

Dialogue · Complete with whose.

A: Do you know the family _____ house was rebuilt after the earthquake?

✓ **whose**

Dialogue · Reorder the sentence.

**predicted / whose / the earthquake / the scientist / research /
won an award**

research won earthquake whose an award scientist the

predicted the

.

✓ **The scientist whose research predicted the earthquake won an award.**

READING

When the Ground Shakes: The Real Cost of a Natural Disaster

Before you read

1. Look at the title. What do you expect this text to compare or describe?
2. What do you already know about the effects of an earthquake on a town or a natural environment?

WHEN THE GROUND SHAKES: THE REAL COST OF A NATURAL DISASTER · 1 / 4

At 4:17 in the morning, the small town of Elmalı, which had never experienced a serious earthquake before, began to shake violently for almost forty seconds. Buildings that had stood for decades collapsed within minutes, and the streets, which were usually silent at that hour, filled with dust, noise and frightened residents searching for their families. By sunrise, rescue teams whose job it was to search the rubble had already pulled dozens of survivors from collapsed buildings, but for many families the search continued for days.

WHEN THE GROUND SHAKES: THE REAL COST OF A NATURAL DISASTER · 2 / 4

The disaster's impact on the surrounding countryside was just as severe as its impact on the town itself. The earthquake triggered several landslides that blocked mountain roads and changed the course of a small river, flooding farmland that had fed the community for generations. Forests on the hillsides, whose roots normally hold the soil in place, were badly damaged, which made the land even more vulnerable to future landslides. Local wildlife, including birds and small mammals that depended on those forests, lost much of their natural habitat almost overnight.

WHEN THE GROUND SHAKES: THE REAL COST OF A NATURAL DISASTER · 3 / 4

In the weeks that followed, thousands of people who had lost their homes were moved to temporary shelters, and this displacement created its own challenges: overcrowding, limited clean water, and the stress of an uncertain future. Yet the relief effort that followed also showed remarkable resilience. Volunteers whose only connection to Elmalı was a desire to help arrived from across the country, and within a month a temporary school, which the community built almost entirely by hand, was already open for local children.

WHEN THE GROUND SHAKES: THE REAL COST OF A NATURAL DISASTER · 4 / 4

Experts who study natural disasters point out that Elmalı's experience is far from unique. Every year, similar events devastate communities that are simply unprepared for what nature can do — a reminder that understanding both the human and the environmental cost of a disaster is the first step towards being ready for the next one.

While you read

What is the main idea of the text?

a

Elmalı's earthquake only damaged buildings, not the countryside.

b

The earthquake had a serious impact on both the town and the natural environment around it.

c

The relief effort in Elmalı failed completely.

While you read

The earthquake triggered several _____ that blocked mountain roads and changed the course of a river.

✓ **landslides**

While you read

According to the text, why did the damaged forests make the land more vulnerable to future landslides?

a

Because the trees attracted more rain

b

Because their roots normally hold the soil in place

c

Because the wildlife left the area

While you read

The temporary school in Elmalı was built entirely by government workers.

a Correct

b Incorrect

✓ The temporary school was built almost entirely by hand by the community and volunteers.

While you read

Thousands of people who had lost their homes experienced _____ , which created overcrowding and limited clean water.

✓ displacement

After you read

1. Which effect of the earthquake — on people or on nature — do you think was more serious? Why?
2. Write one sentence describing how you think you would feel in Elmalı's situation, using a relative clause.

READING

Can Technology Predict the Next Disaster?

Before you read

1. What do you think technology can and cannot do to prevent damage from a natural disaster?
2. Have you ever heard of an app or a system that warns people about floods, earthquakes or wildfires?

CAN TECHNOLOGY PREDICT THE NEXT DISASTER? · 1 / 4

For most of human history, natural disasters arrived almost without warning. Today, however, artificial intelligence and data analysis are changing that completely. Scientists who study climate patterns now use algorithms that can process huge amounts of satellite data, weather records and historical information — far more than any human team could analyse alone — to identify the early signs of an earthquake, flood or wildfire before it happens.

CAN TECHNOLOGY PREDICT THE NEXT DISASTER? · 2 / 4

One clear example is the early warning system that many coastal cities have started to install. Sensors, which are placed along riverbanks and coastlines, constantly send data to a central system whose algorithm compares current conditions with thousands of past events. When the pattern matches a known danger, the system can send an alert to residents' phones minutes — or sometimes hours — before a flood or a tsunami arrives. In regions prone to wildfires, similar technology uses satellite images to detect the first signs of smoke, often long before anyone on the ground notices anything unusual.

CAN TECHNOLOGY PREDICT THE NEXT DISASTER? · 3 / 4

The benefits of this technology are not limited to prediction. After a disaster strikes, hospitals whose emergency departments would otherwise be overwhelmed can use data models to estimate how many patients they are likely to receive and what kind of injuries to expect, allowing them to prepare staff and supplies more effectively. Humanitarian organisations, which used to rely mainly on phone calls and paper reports, now use real-time data to decide where relief teams are needed most urgently.

CAN TECHNOLOGY PREDICT THE NEXT DISASTER? · 4 / 4

Of course, technology is not a perfect solution. Systems that depend on expensive sensors and reliable internet connections are often unavailable in the poorest and most vulnerable communities — the very places that frequently need them most. Experts agree that AI and data can dramatically reduce the damage a disaster causes, but only if the people who design these systems remember to make them available to everyone, not just to wealthy cities that can already afford them.

While you read

What is the main purpose of the text?

a To explain why technology cannot help with natural disasters

b To describe how AI and data are changing disaster prediction and response

c To list every country that has an early warning system

While you read

Sensors placed along riverbanks and coastlines send data to a central system whose _____ compares current conditions with past events.

✓ **algorithm**

While you read

How can data models help hospitals after a disaster?

a

By replacing doctors completely

b

By estimating patient numbers and likely injuries so staff can prepare

c

By predicting exactly which patients will survive

While you read

According to the text, early warning technology is equally available to every community in the world.

a Correct

b Incorrect

✓ Early warning technology is often unavailable in the poorest and most vulnerable communities.

While you read

Humanitarian organisations now use _____ data to decide where relief teams are needed most urgently.

✓ **real-time**

After you read

1. Do you agree that AI can 'dramatically reduce the damage a disaster causes'? Why or why not?
2. What is one problem with relying on technology for disaster prediction, according to the text?

LISTENING

After the Flood: A Radio Interview

Before you listen

1. Predict: what do you think the rescue involved, and what does Deniz think about technology and disasters?

After the Flood: A Radio Interview

Ses kaydı yakında — transkripti öğretmen okuyabilir.

While you listen

What method did Deniz's team mainly use for the rescue?

a Helicopters and drones

b Small boats and teamwork

c Government trucks

Give 10 seconds to think. Students show 1, 2 or 3 fingers for a, b, c. Then reveal. / 10 sn düşünme süresi; öğrenciler parmakla oy verir, sonra cevabı açın.

While you listen

According to Deniz, the local disaster response _____
worked well.

✓ system

While you listen

Deniz thinks technology could never help prevent this kind of disaster.

a Correct

b Incorrect

✓ **Deniz thinks technology and early warnings could help prevent future disasters.**

While you listen

In the word 'guest', the letter 'u' is:

a pronounced /juː/

b silent

c pronounced /ʌ/

LISTENING

Predicting the Unpredictable — A Project Brief

Before you listen

1. What natural disaster and technology idea would you choose to research if you did this project?

Predicting the Unpredictable — A Project Brief

Ses kaydı yakında — transkripti öğretmen okuyabilir.

While you listen

What are students asked to research?

a A famous scientist

b A real natural disaster
and a technology
solution

c The history of their
school

While you listen

Students should find out about the disaster's environmental and _____ impact.

✓ **human**

While you listen

Students can only present individually.

a Correct

b Incorrect

✓ **Students can present individually or in pairs.**

SPEAKING

Nature Under Threat: Discussing Disaster Impact

In pairs, discuss the environmental and human impact of a natural disaster you know about (from the news, Reading 1, or your own experience). Use at least three relative clauses (defining, non-defining or whose) to give extra information, and try to use topic vocabulary accurately.

SPEAKING

My Tech Solution for Disasters — A Mini-Presentation

Prepare a two-minute talk presenting a technology idea (AI, an app, a data system) that could help predict or reduce the impact of a natural disaster. Explain the problem, your solution, and why it would make a difference.

WRITING

A Report: A Natural Disaster and a Tech Solution

Write a short report for your class project. Describe a real or realistic natural disaster, its impact on nature and people, and suggest one technological solution that could reduce its effects. Use at least two relative clauses.

WRITING

Opinion Blog Post: Can AI Really Solve Global Problems?

Write a blog post giving your opinion on whether technology such as AI can really help solve global problems like natural disasters and healthcare. Support your opinion with at least two reasons and one example.

REVIEW

Exit ticket

Exit ticket

The scientists _____ study earthquakes work at the research centre.

a which

b who

c whose

Exit ticket

The flood, _____ destroyed hundreds of homes, happened last spring.

✓ which

Exit ticket

This is the app _____ predicts wildfires using AI.

a who

b which

c whose

Exit ticket

The engineer _____ app won the innovation prize studied in Istanbul.

✓ whose

Exit ticket

Which sentence is correctly punctuated?

a My uncle who works for
a relief organisation is
here.

b My uncle, who works
for a relief
organisation, is here.

c My uncle, who works
for a relief organisation
is here.

Exit ticket

The report we read yesterday, was alarming.

a Correct

b Incorrect

✓ The report we read yesterday was alarming.

Exit ticket

The data the sensors _____ (collect) helps predict floods.

✓ collect

Exit ticket

The village _____ water supply was affected by the drought needs help.

a who

b whose

c which

Exit ticket

Plastic bags, _____ take hundreds of years to decompose, harm marine life.

a who

b which

c whose

Exit ticket

The volunteers _____ clean the beach every month are mostly students.

✓ who / that

Exit ticket

Put the words in order.

which

recycling

bin,

to

needed

full,

was

be

the

emptied

•

✓ The recycling bin, which was full, needed to be emptied.

Exit ticket

The polar bear that live in the Arctic, is endangered.

a Correct

b Incorrect

✓ **The polar bear that lives in the Arctic is endangered.**

Exit ticket

Rewrite the two sentences as one, using a relative clause with 'whose'.

(The factory pollutes the river. Its chemicals are toxic.)

✓ **The factory whose chemicals are toxic pollutes the river.**

Exit ticket

The rainforest, _____ produces a huge amount of oxygen, is being destroyed rapidly.

✓ which